by a bolt of lightning. But higher in the atmosphere, where the sunlight reaches, a much more beautiful vista awaits.

On Jupiter, above a high-altitude haze layer composed of ammonia (rather than water) ice particles, the sky is almost black. Farther down, in the blue sky region, are multicolored clouds—in various shades of yellow-brown, and of unknown composition. (The candidate materials include sulfur, phosphorus, and complex organic molecules.) Even farther down, the sky will appear red-brown, except that the clouds there are of varying thicknesses, and where they are thin, you might see a patch of blue. Still deeper, we gradually return to perpetual night. Something similar is true on Saturn, but the colors there are much more muted.

Uranus and especially Neptune have an uncanny, austere blue color through which clouds—some of them a little whiter—are carried by high-speed winds. Sunlight reaches a comparatively clean atmosphere composed mainly of hydrogen and helium but also rich in methane. Long paths of methane absorb yellow and especially red light and let the green and blue filter through. A thin hydrocarbon haze removes a little blue. There may be a depth where the sky is greenish.

Conventional wisdom holds that the absorption by methane and the Rayleigh scattering of sunlight by the deep atmosphere together account for the blue colors on Uranus and Neptune. But analysis of *Voyager* data by Kevin Baines of JPL seems to show that these causes are insufficient. Apparently very deep—maybe in the vicinity of hypothesized clouds of hydrogen sulfide—there is an abundant blue substance. So far no one has been able to figure out what it might be. Blue materials are very rare in Nature. As always happens in science, the old mysteries are dispelled only to be replaced by new ones. Sooner or later we'll find out the answer to this one, too.

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